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PREVENTION AND EARLY DETECTION

Fecal Hemoglobin Levels in Prior Negative Screening and Detection of Colorectal Neoplasia: A Dose-Response Meta-Analysis

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BACKGROUND & AIMS: Prior studies have shown that individuals with fecal hemoglobin (f-Hb) concentrations just below the positivity cutoff have an increased colorectal cancer risk compared with those with no or low f-Hb. Understanding the dose-response association between f-Hb in a prior screening round and the detection of colorectal neoplasia is crucial for tailoring risk-based screening recommendations. **METHODS:** We searched the literature to identify studies reporting the association between f-Hb in prior screening round and colorectal neoplasia detection in an average-risk population. Analysis involved a 2-stage approach using log-log regression models to assess dose-response relationships across studies, with effect sizes pooled using a random effects model. Heterogeneity was assessed by excluding individual studies in sensitivity analyses. Subgroup analyses examined variations in effects by outcome definitions and detection methods. **RESULTS:** This systematic review and meta-analysis included 13 studies with 4,493,672 individuals. All studies demonstrated a positive association between f-Hb in prior screenings and colorectal neoplasia detection. Pooled analysis revealed that individuals with f-Hb concentrations of 5, 10, 20, and 40 μg/g had a 3-, 5-, 8-, and 13-fold higher risk of colorectal neoplasia, respectively, compared with individuals with 0 μg/g. Although significant heterogeneity ($I^2 = 97.5\%$, $P < .001$) was observed, sensitivity analyses confirmed the consistency of findings. Subgroup analyses indicated that f-Hb concentrations from previous negative tests were especially predictive of advanced neoplasia in subsequent screenings. **CONCLUSIONS:** Our findings suggest that the risk of detecting colorectal neoplasia increases with prior f-Hb concentrations in negative tests, supporting the development of risk-stratified screening strategies based on these concentrations.

Keywords: Fecal Hemoglobin; Colorectal Neoplasia; Dose-Response Meta-Analysis; Risk-Based Screening.

Colorectal cancer (CRC) is the second most common cause of cancer death world-wide, with an estimated 935,173 deaths (9.4% of all cancer deaths) in 2020.[1] Fortunately, CRC screening has been shown to reduce CRC-related incidence and mortality.[2] The fecal immunochemical test (FIT) is the primary screening test for CRC screening in many countries.[3] In most FIT-based screening programs, individuals are referred for a colonoscopy if the fecal hemoglobin concentration (f-Hb) exceeds a prespecified positivity cutoff. All individuals with a FIT value below the cutoff are re-invited for FIT screening after a fixed interval (2 years in most programs). This means that FIT is currently used in a binary manner, overlooking its potentially valuable quantitative data. Nonetheless, prior studies have shown that individuals with an f-Hb value just below the cutoff have an increased CRC risk compared with individuals with no f-Hb or very low concentrations.[4–22]

Given the established association between f-Hb and CRC risk, risk-stratifying CRC screening programs based on quantitative FIT values could potentially improve the harm-benefit ratio of CRC screening. Accurate risk assessment plays a crucial role in tailoring screening strategies to individuals, as misclassification of individuals into risk groups can lead to 2 potential issues: over- or underscreening. Overscreening could lead to more false-positives and overdiagnosis, whereas underscreening could delay the detection of colorectal neoplasia.

Among the published studies, a categorical approach was used to examine the association between categories of f-Hb concentration in the prior screening round and detection of colorectal neoplasia. Due to the global variation in FIT positivity cutoffs and the variation in defining f-Hb categories, as reported in published studies, comparisons across different studies pose challenges. Furthermore, obtaining a single effect measure for the association is beneficial for risk stratification in CRC screening, but a dose-response association has yet to be established. To address this, we conducted a systematic review and dose-response meta-analyses to provide more insight into the quantitative, dose-response association between prior f-Hb concentrations and the detection of colorectal neoplasia.

Methods

Search Strategy and Selection Criteria

For this systematic review and meta-analysis, an initial literature search was conducted in Embase, Medline, Cochrane, Web of

Abbreviations used in this paper: **AN, advanced neoplasia; CI, confidence interval; CRC, colorectal cancer; f-Hb, fecal hemoglobin; FIT, fecal immunochemical test; HR, hazard ratio; OR, odds ratio; RR, relative risk.**

Most current article

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WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

Colorectal cancer screening is essential for early detection and prevention. Fecal immunochemical tests, which measure fecal hemoglobin, are widely used. Research suggests that individuals with fecal hemoglobin levels just below the positivity threshold may still face a higher risk of colorectal cancer compared with those with very low or undetectable levels.

NEW FINDINGS

Individuals with fecal hemoglobin concentrations of 5, 10, 20, and 40 μg/g had a 3-, 5-, 8-, and 13-times higher risk of colorectal neoplasia detection in the next screening round compared with those with 0 μg/g.

LIMITATIONS

There was considerable heterogeneity among the included studies, such as in detection methods, outcome definitions, positivity thresholds, prior screening histories, and the ages at which screening started and ended. This variability may have affected the results.

BASIC RESEARCH RELEVANCE

This study enhances the understanding of how fecal hemoglobin levels from a prior screening round correlate with future colorectal cancer risk.

CLINICAL RESEARCH RELEVANCE

These findings highlight the potential for tailoring colorectal cancer screening strategies based on fecal hemoglobin levels from previous negative fecal immunochemical test results. This risk-stratified approach could improve early detection and optimize the use of screening resources.

Science, and Google Scholar databases to identify all eligible studies published up to August 8, 2023, on the dose-response association between f-Hb in the previous screening round and detection of colorectal neoplasia. The databases were searched using a combination of terms related to hemoglobin level (eg, "occult blood," "fecal hemoglobin"), colorectal neoplasia (eg, "colorectal cancer," "advanced neoplasia," "colon lesion"), predictive value (eg, "prediction," "risk," "predictor") and prior FIT (eg, "negative FIT," "result below threshold," "previous FIT"). Supplementary Appendix 1 shows the full search strategy for each database.

Studies were included within the systematic review and meta-analysis if they met the following criteria: (1) studies that had the association between f-Hb concentration in prior screening round and detection of colorectal neoplasia as an outcome, (2) studies that reported outcomes for the average-risk population, (3) studies that reported relative risk (RR), hazard ratios (HRs), or odds ratios (ORs) and 95% confidence intervals (CIs) or absolute numbers from which RRs and CIs could be calculated.

After removal of duplicates, all eligible studies were screened based on title and abstract. If the abstract did not contain adequate details for evaluating alignment with eligibility criteria, a full-text screening was carried out. Subsequently, all remaining studies were screened based on the full text. If the study cohorts overlapped (ie, FIT results from the same region with similar time periods of data collection), we initially excluded the conference abstracts. If overlap persisted, we then selected the article with the highest number of colorectal neoplasia cases. Both title/abstract screening as well as full-text screening were carried out independently by 2 authors (D.M.N.v.d.B. and R.v.d.P.) to reduce bias. A third author was consulted (E.T.-Z.) if there was a discussion about the eligibility of a study. This systematic review and meta-analysis was prospectively registered with the PROSPERO database (CRD42022371150).

Data Extraction

The following data were extracted from each study: name of the first author, year of publication, country in which the study was performed, study setting, study period, FIT positivity cut-off, screening interval for studies with multiple screening rounds or follow-up duration for studies with once-only screening, FIT brand, number of participants, f-Hb categories and the number of individuals in each f-Hb category, number of detected cases, age range, mean or median age at prior FIT, sex distribution, definition of outcome measure, adjustment for covariates, RR/OR/HR and their corresponding 95% CI. In studies in which data from 2 preceding screening rounds were available, we looked at the f-Hb categories and RR estimates of the most recent screening round. If solely the aggregated f-Hb concentration from 2 rounds was available, we divided the f-Hb concentrations by 2 to obtain the average f-Hb categories for 1 round.

The outcome measure in this study, colorectal neoplasia, included "any colorectal neoplasia," "colorectal cancer," "advanced neoplasia," and/or "high-risk neoplasia." Because the prevalence of colorectal neoplasia in individuals with a prior negative FIT is relatively low, we assumed that both HRs and ORs were comparable estimates of RR.[23] For studies in which RRs were not available, we computed the unadjusted RRs and CIs using the raw data presented in the paper.

Data Analysis

A 2-stage procedure was used to obtain the pooled dose-response association of all included studies. First, study-specific trends were obtained using linear regression. For each study, the mean of the upper and lower boundaries of each f-Hb category was assigned to the corresponding RR estimate, except for the first category. For the first category, we assumed that most individuals had no blood in their stool, so the f-Hb value was set to 0. To avoid running into negative predicted RRs after fitting linear regression, the response variable was log-transformed. An examination of the included studies suggested a concave dose-response curve, in which the greatest difference in RR occurs between individuals with 0 and 10 μg f-Hb. For higher f-Hb concentrations, the difference in RR appeared to flatten out. Therefore, a log-log regression model was selected, meaning that both the response variable and the explanatory variable were log-transformed.

In the second stage, the study-specific effect sizes were pooled using a random effects model. The implementation of the random effects model was conducted using the R package *dosresmeta*.[24] Each individual study was weighted by the inverse of its variance. To estimate the variance, the

Greenland and Longnecker method was used.[25] This method considers that the log RRs for the different f-Hb categories of a single study are correlated given that they are all expressed in relation to a reference category. To estimate the correlations, this method requires information on the sample size and number of cases in each f-Hb category. For studies missing sample size and/or number of cases, we assumed no correlation between categories.

The extent of heterogeneity in RRs across studies was assessed using the Cochrane Q-statistic[26] (a significance level of $P < .10$ was suggestive of significant heterogeneity) and the I^2-statistic[27] (25%, 50%, and 75% were categorized as low, moderate, and high heterogeneity, respectively). To investigate potential sources of between-study heterogeneity, we performed sensitivity analyses. First, we conducted a sensitivity analysis in which we systematically excluded 1 study at a time to identify the study that exerted the most significant influence on the pooled effect size. Second, we performed a sensitivity analysis in which we excluded all conference abstracts. Third, we conducted a sensitivity analysis in which studies not reporting the sample size and number of cases in each f-Hb category were excluded. Furthermore, we investigated the change in the pooled estimate if we included only studies that adjusted for covariates. Finally, we performed subgroup analyses to identify the effects of the outcome measure definition and the method of detection. For the outcome measure definition, we assessed any colorectal neoplasia alongside (1) advanced neoplasia (AN), (2) CRC, (3) advanced adenoma, and (4) advanced adenoma combined with early-stage CRC. AN included all studies that categorized their outcome measure as either "advanced neoplasia" or "high-risk neoplasia." For the detection method, we examined outcomes for the combination of screen-detected and clinically detected lesions, screen-detected lesions only, and clinically detected lesions only.

Results

Included Studies

Figure 1 shows the flowchart of the study selection process. A total of 1333 records were identified from the databases Embase (595 records), Medline (283 records), Web of Science (286 records), Cochrane (45 records), and Google Scholar (124 records). After removing duplicates, 784 records were screened by title and abstract. Based on title and/or abstract, 40 records were assessed for eligibility by reading their full text (Supplementary Table 1). This resulted in 13 eligible studies for data extraction.

The characteristics of the included studies are summarized in Table 1. The meta-analysis included a large sample of 4,493,672 individuals from 13 studies. The studies originated from Spain, France, the Netherlands, Taiwan, Denmark, Scotland, Ireland, Korea, Italy, and Norway. Most studies reported data from a cohort study design,[5,6,8,10–12,14–18,22] and 1 study[20] reported data from a randomized controlled trial. Most studies followed individuals with a negative FIT until the subsequent screening, equivalent to a 2-year

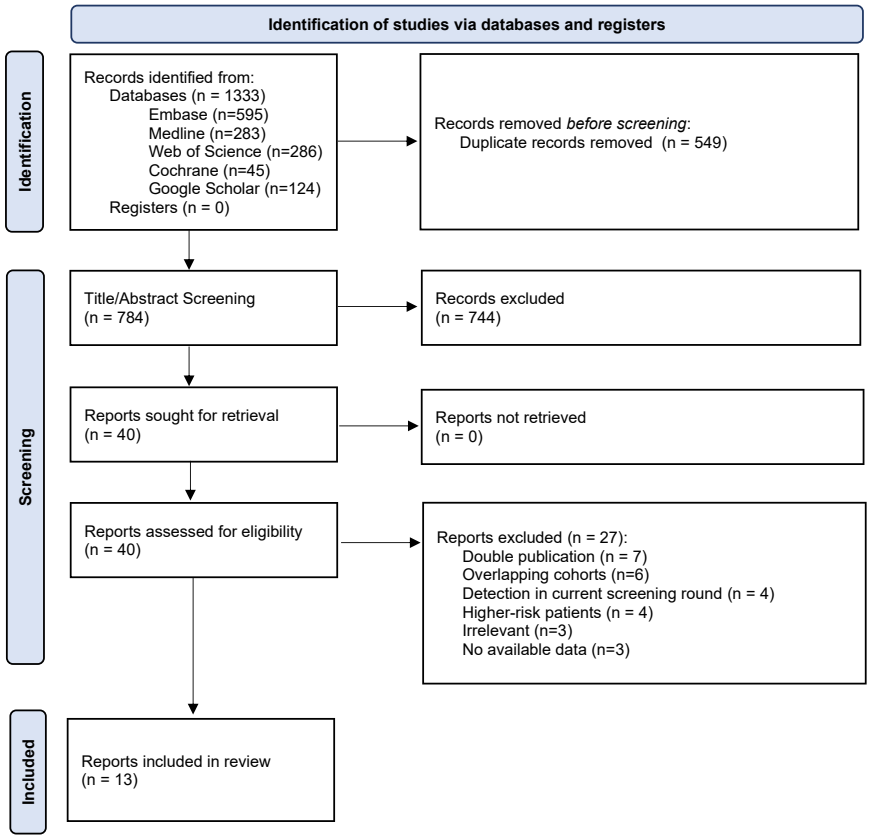


Figure 1. Flowchart of the study selection process.

Table 1. Characteristics of the Studies Included in the Meta-Analysis

First author, publication year, country	Study setting, Study period	Follow-up duration	Number of participants with a negative prior FIT, age range, mean/ median age at prior FIT, sex distribution	Measurement of exposure, FIT brand, cutoff	Outcome, SD or CL	Exposure categories (in μg f-Hb/g)	RR, OR, or HR (95% CI)	Number of individuals in each f-Hb category (%)	Adjustment for covariates
Balamou, C., 2021, France	French CRC screening program, 2015–2019	Minimum of 1 year between negative and positive test. Most had an interval >2 y.	293,750 participants, 50–74 y, mean age: 61, 45% male	f-Hb in previous round, OC-Sensor, 30 μg f-Hb/g	8748 cases of any colorectal neoplasia, SD	0 0.1–10 10.1–20 20.1–29.9	1.0 3.1 (2.6–3.7) 4.4 (4.2–4.7) 6.6 (6.1–7.2)	280,012 (95.3) 1,680 (0.6) 8,776 (3.0) 3,282 (1.1)	No. Calculated from raw data.[a]
Breekveldt, E.C.H., 2023, the Netherlands	Dutch CRC screening program, 2014–2017	2 y	2,153,582 participants, 55–75 y, median age: 67, 48% male	f-Hb in first screening round, FOB-Gold, 47 μg f-Hb/g	2256 cases of CRC, CL	0–2.6 2.7–10 10.1–20 20.1–30 30.1–40 40.1–46.9	1.0 4.0 (3.5–4.5) 7.2 (6.3–8.1) 11.1 (9.5–12.9) 14.9 (12.7–17.4) 16.9 (13.9–20.3)	1,907,528 (88.7) 127,256 (5.9) 62,479 (2.9) 26,723 (1.2) 18,603 (0.9) 10,993 (0.5)	Sex, age.
Burón, A., Román, M., 2019, Spain[b]	Barcelona CRC screening program, 2012–2015	2 y	42,524 participants, 50–69 y, median age: 57–61, 43% male	f-Hb in round 2, OC-Sensor, 20 μg f-Hb/g	1078 cases of any colorectal neoplasia, SD or CL	0–3.8 3.9–10 10.1–19.9	1.0 3.5 (3.0, 4.1) 5.8 (4.9, 6.9)	38,831 (91.3) 2,470 (5.8) 1,223 (2.9)	No. Calculated from raw data.[a]
Chen, L. S., 2011, Taiwan	Keelung community-based CRC screening program, 2001–2007	Median follow-up duration of 4.4 y	44,324 participants, 40–69 y, mean age: 52, 40% male	f-Hb in previous screening round, OC-Sensor, 20 μg f-Hb/g	400 cases of any colorectal neoplasia, SD or CL	0 1–3.9 4–7.9 8–11.9 12–15.9 16–19.9	1.0 1.2 (1.0–1.5) 2.4 (1.9–3.0) 3.3 (2.4–4.6) 3.3 (2.0–5.2) 5.5 (3.4–9.0)	18,995 (42.9) 15,644 (35.3) 6,216 (14.0) 1,998 (4.5) 963 (2.2) 508 (1.1)	No. Calculated from raw data.[a]
Deding, U., 2023, Denmark	Danish CRC Screening Program, 2014–2018	2 y	1,160,902 participants, 50–74 y, median age: 60–64, 45% male	f-Hb in round 1, OC-Sensor, 20 μg f-Hb/g	829 cases of CRC, CL	0–7.1 7.2–11.8 11.9–19.8	1.0 3.4 (2.8–4.0) 7.1 (5.9–8.6)	1,061,991 (91.5) 70,375 (6.1) 28,536 (2.5)	Sex, age, educational level, income quartiles.

Table 1. Continued

First author, publication year, country	Study setting, Study period	Follow-up duration	Number of participants with a negative prior FIT, age range, mean/ median age at prior FIT, sex distribution	Measurement of exposure, FIT brand, cutoff	Outcome, SD or CL	Exposure categories (in μg f-Hb/g)	RR, OR, or HR (95% CI)	Number of individuals in each f-Hb category (%)	Adjustment for covariates
Digby, J., 2017, Scotland[b]	Scottish CRC Screening Program, 2011–2015	2 y	30,618 participants, 50–74 y , NA, NA	f-Hb in previous round, OC-Sensor, 80 μg f-Hb/g	111 cases of any colorectal neoplasia, SD	0–19.9 20–39.9 40–59.9 60.1–79.9	1.0 10.8 (7.6, 15.5) 12.5 (7.2, 21.8) 21.6 (12.5, 37.2)	29,498 (96.3) 720 (2.4) 253 (0.8) 147 (0.5)	No. Calculated from raw data.[a]
Digby, J., 2023, Scotland	Scottish CRC Screening Program, 2017–2021	2 y	224,541 participants, 50–74 y, median age: 60–64, 48% male	f-Hb in previous round, OC-Sensor, 80 μg f-Hb/g	141 cases of CRC, CL	0–9.9 10–19.9 20–39.9 40–79.9	1.0 6.2 (3.8–10.1) 11.7 (7.6–18.0) 14.3 (8.9–22.9)	202,233 (90.0) 10,076 (4.5) 7,418 (3.3) 4,814 (2.1)	No.
Gibson, D. J., 2019, Ireland[b]	Ireland's CRC screening program, 2012–2017	2 y	37,877 participants, 60–70 y, median age: 67, 46% male	f-Hb in first round, OC-Sensor, 15 & 45 μg f-Hb/g	318 cases of any colorectal neoplasia, SD	0–5 5.1–10 10.1–15 15.1–20 20.1–25 25.1–30 30.1–35 35.1–40 40.1–44.9	1.0 3.1 (2.3, 4.1) 5.8 (4.2, 8.1) 10.5 (7.3, 15.0) 11.2 (6.8, 18.5) 17.6 (11.3, 27.6) 15.0 (8.8, 25.5) 10.9 (5.4, 21.9) 22.5 (13.3, 38.1)	29,943 (79.1) 4,325 (11.4) 1,652 (4.4) 755 (2.0) 354 (0.9) 281 (0.7) 231 (0.6) 182 (0.5) 154 (0.4)	No. Calculated from raw data.[a]
Grobbee, E. J., 2017, the Netherlands	Dutch population-based cohort, 2006–2014	Median follow-up duration of 4.7 y	7663 participants, 50–74 y, median age: 58, 47% male	f-Hb at first screening, OC-Sensor and FOB-Gold, 10 μg f-Hb/g	221 cases of advanced neoplasia (size $\geq$10 mm, >25% villous histology and/ or high-grade dysplasia or CRC), SD or CL	0 0.1–2 2.1–4 4.1–6 6.1–8 8.1–9.9	1.0 1.2 (0.9–1.7) 2.8 (1.7–4.4) 5.7 (3.7–8.9) 4.2 (2.1–8.7) 8.2 (4.5–15.0)	4927 (64.5) 1874 (24.5) 436 (5.7) 214 (2.8) 106 (1.4) 78 (1.0) (subjects at risk 1 year after FIT)	Sex, age and socioeconomic status.
Kim, S. Y., 2019, Korea	Korean population-based cohort, 2009–2016	5 y	49,465 participants, NA, NA, NA	f-Hb at baseline screening, NA, 20 μg f-Hb/g	140 cases of CRC, detected with colonoscopy within 5 y of FIT	0–5 5.1–10 10.1–15 15.1–19.9	1.0 1.9 (1.1–3.4) 3.1 (1.3–7.5) 4.4 (1.6–11.8)	NA	Adjusted HRs reported, but no covariates reported.

Table 1. Continued

First author, publication year, country	Study setting, Study period	Follow-up duration	Number of participants with a negative prior FIT, age range, mean/ median age at prior FIT, sex distribution	Measurement of exposure, FIT brand, cutoff	Outcome, SD or CL	Exposure categories (in μg f-Hb/g)	RR, OR, or HR (95% CI)	Number of individuals in each f-Hb category (%)	Adjustment for covariates
Meester, R. G. S., 2022, the Netherlands[b]	Dutch CRC screening program, 2014–2018	2 y	265,881 participants, 64–72 y, mean age: 69, 48% male	f-Hb in 2nd round, FOB-Gold, 47 μg f-Hb/g	3254 cases of advanced neoplasia (size $\geq$10 mm, >25% villous histology and/ or high-grade dysplasia or CRC) in 3rd round, SD	0 0.1–9.9 10–19.9 20–29.9 30.-39.9 40–46.9	1.0 4.8 (4.3–5.3) 6.2 (5.5–7.0) 7.9 (6.8–9.1) 8.0 (6.7–9.5) 8.6 (7.0–10.5)	242,220 (91.1) 5322 (2.0) 8253 (3.1) 4887 (1.8) 2362 (0.9) 1742 (0.7) 1095 (0.4)	Sex, age and f-Hb concentration in the first screening round.
Ribe, S., 2023, Norway[b]	CRC screening trial, 2012–2020	2 y	18,522 participants, 50–76 y, median age: 63, 47% male	f-Hb in 1st round, OC-Sensor, 15 μg f-Hb/g	245 cases of advanced neoplasia (CRC or size $\geq$10 mm, high-grade dysplasia or villous/ tubulovillous histology) in 2nd round, SD	0 0.1–5 5.1–14.9	1.0 1.5 (1.1–2.0) 5.5 (3.9–7.6)	9,503 (51.3) 7,576 (40.9) 1,443 (7.8)	Sex, age, time interval between FITs, center.
Senore, C. 2020, Italy[b]	4 population-based screening programs in Northern-Central Italy, 2004–2014	4 y	164,023 participants, 50–69 y, NA, NA	Average f-Hb of the first 2 rounds, OC-Sensor, 20 μg f-Hb/g	1436 cases of advanced neoplasia (size $\geq$10 mm, high-grade dysplasia or villous component >20% or CRC), SD	0 0.1–1.9 2–4.9 5–7.4 7.5–9.9 10–19.9	1.0 1.8 (1.6–2.1) 4.6 (3.9–5.4) 9.3 (7.7–11.2) 12.4 (10.4–15.8) 32.5 (26.2–40.4)	80 579 (49.1) 54 352 (33.1) 19 715 (12.0) 5336 (3.3) 2912 (1.8) 1129 (0.7)	Age and interval since last FIT.

CL, clinically detected; NA, not available; SD, screen-detected.
[a]The RR is calculated from the raw data. See Supplementary Appendix 2 for the calculations.
[b]This study provided data on several outcome measures, such as advanced adenoma, advanced neoplasia, and CRC. In the main analysis, we reported the RRs for all these lesions combined. Additional RRs for other outcome measures are available in Supplementary Appendix 2.

interval; however, some studies deviated from the 2-year follow-up. The study by Balamou et al[5] included a range of follow-up times from 1 to 4 years, with most participants having a follow-up of at least 2 years. The study by Senore et al,[22] which encompassed 2 negative screening rounds, reported a follow-up period of 4 years. Furthermore, Chen et al,[10] Grobbee et al,[16] and Kim et al[17] reported a median follow-up of 4.4, 4.7, and 5 years, respectively.

Almost all studies reported the f-Hb concentration measured in the prior screening round, whereas 1 study combined the f-Hb concentration of 2 previous screening rounds by using the cumulative f-Hb value.[22] There was a large variability in the positivity cutoffs reported in the included studies, ranging from 10 μg f-Hb/g to 80 μg f-Hb/g. The most common positivity cutoff was 20 μg f-Hb/g. Two different FIT brands were reported: the OC-Sensor (Eiken Chemical Co, Tokyo, Japan) and the FOB-Gold (Sentinel, Milan, Italy). The OC-Sensor was used in most of the studies, whereas the FOB-Gold was used only in Dutch studies.[6,16,18]

There was significant heterogeneity in the definition of the outcome variable between studies. Although some studies included any colorectal neoplasia, others solely focused on AN or CRC. There was also considerable variation in the definition of AN. The most commonly used definition was cancer or an adenoma of at least 10 mm in diameter, or with high-grade dysplasia, or with villous or tubulovillous histologic characteristics, or any combination thereof.[16,18,20,22] Digby et al[12] used cancer or adenomas of at least 10 mm in diameter, or at least 3 adenomas. Six studies[5,12,15,18,20,22] reported on screen-detected colorectal neoplasia only, 3 studies[6,11,14] on clinically detected colorectal neoplasia, 3 studies[8,10,16] on both screen- and clinically detected, and one study[17] did not report on method of detection. Most studies adjusted for potential confounders such as age and sex.

Dose-Response Association

All 13 included studies showed a significant positive association (ie, log RR >0) of f-Hb in the prior screening round and the detection of colorectal neoplasia (Figure 2 and Supplementary Figure 1). The study with the largest estimated effect size was the study by Senore et al,[22] with an estimated effect size of 1.20. Kim et al[17] had the smallest estimated effect size of 0.40. Pooling the individual curves resulted in a pooled effect size of 0.69 (95% CI, 0.58–0.79) (Figure 2). Using an f-Hb concentration of 0 μg/g as the reference level, the predicted RR for an f-Hb concentration of 5, 10, 15, 20, and 40 μg/g was 3.4, 5.2, 6.7, 8.1, and 12.8 (Table 2 and Figure 3). Notably, the largest increase in RR occurred between 0 and 10 μg/g, after which the incremental increase gradually decreased.

A high heterogeneity was identified, with an I^2 of 97.5% and a Cochrane Q-statistic of 487 ($P < .001$). Substantial between-study heterogeneity led to relatively similar weighting of individual studies. This heterogeneity has been considered in the prediction interval, which provides the range of effect sizes where it is expected that 95% of future studies will lie (Figure 3).

Excluding 1 study at a time resulted in a pooled effect size ranging from 0.64 to 0.71 (Supplementary Figure 2). The largest deviation from the original pooled estimate occurred when the study by Senore et al[22] was excluded, leading to a pooled effect size of 0.64 (95% CI, 0.59–0.70; $I^2 = 93.2\%$). The meta-analysis included a conference abstract by Kim et al,[17] and removing this study produced a pooled effect size of 0.71 (95% CI, 0.60–0.81; $I^2 = 97.7$). Studies by Grobbee et al[16] and Kim et al[17] lacked reporting on sample sizes and/or the number of cases for each f-Hb category. Excluding these studies yielded a pooled effect size of 0.70 (95% CI, 0.59–0.81; $I^2 = 97.9$). When only studies with adjusted RRs were analyzed, the pooled effect size was 0.73 (95% CI, 0.55–0.91; I^2=98.0).

Our findings demonstrated a clear trend where increasing f-Hb concentrations were associated with higher RRs for various types of colorectal neoplasia, but the degree of risk varied among the outcome definition (Table 2, Supplementary Figures 3 and 4). At a lower f-Hb concentration of 5 μg/g,

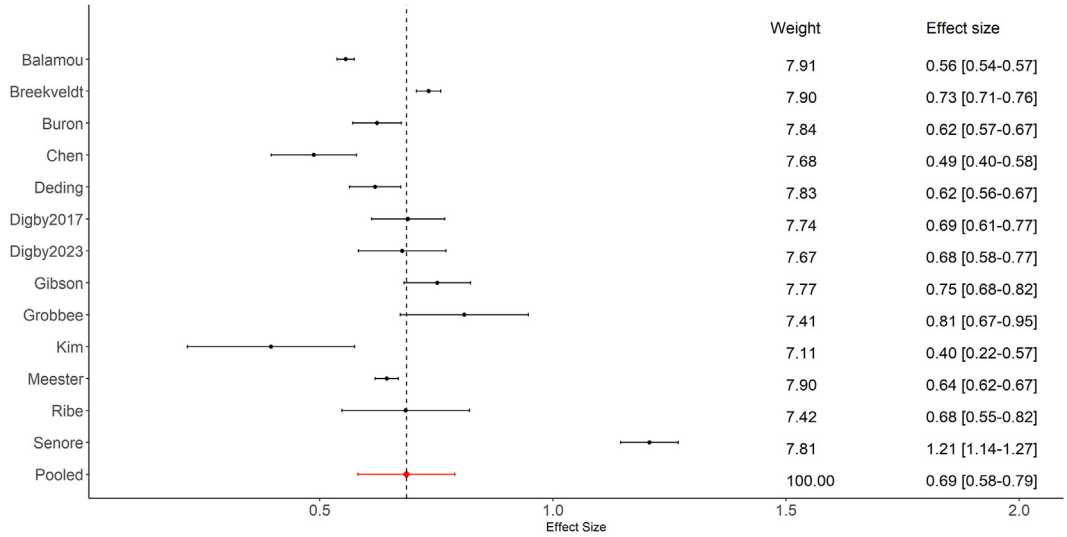


Figure 2. Forest plot of the effect size with corresponding 95% CI for the association between log f-Hb and log RR of detecting colorectal neoplasia.

Table 2. Predicted RR of Detecting Colorectal Neoplasia, Advanced Neoplasia, CRC, Advanced Adenomas, and Advanced Adenomas + Early-Stage CRC Based on Prior f-Hb Concentration

	Pooled RR (95% CI)				
Prior f-Hb concentration	Any colorectal neoplasia (all 13 studies)	Advanced neoplasia	CRC	Advanced adenoma	Advanced adenoma + early-stage CRC
0	1.0	1.0	1.0	1.0	1.0
5	3.4 (2.8–4.1)	4.3 (3.5–5.3)	3.8 (3.0–4.8)	4.5 (3.3–6.1)	4.4 (2.9–6.8)
10	5.2 (4.0–6.6)	7.0 (5.3–9.3)	6.1 (4.4–8.2)	7.5 (5.0–11.2)	7.3 (4.1–13.0)
15	6.7 (5.0–8.9)	9.5 (6.8–13.1)	8.0 (5.6–11.5)	10.3 (6.5–16.4)	11.0 (5.2–19.4)
20	8.1 (5.9–11.1)	11.8 (8.3–16.9)	9.8 (6.6–14.6)	12.9 (7.7–21.6)	12.5 (6.1–26.0)
40	12.8 (8.7–18.8)	20.3 (13.2–31.4)	16.2 (10.1–26.2)	22.7 (12.1–42.4)	21.9 (9.0–53.1)

the RR for any colorectal neoplasia was 3.4, whereas it was higher for AN at 4.3, CRC at 3.8, advanced adenomas at 4.5, and advanced adenomas combined with early-stage CRC at 4.4. As f-Hb concentrations increased, the differences among these categories became more pronounced. For instance, at 20 μg/g, the RR was 8.1 for any colorectal neoplasia, but it increased to 11.8 for AN, 9.8 for CRC, 12.9 for advanced adenomas, and 12.5 for advanced adenomas and early-stage CRC. However, the differences between these categories were not statistically significant. Moving from any colorectal neoplasia to more specific subgroups did not consistently reduce heterogeneity (Supplementary Table 2).

No significant differences in risk were observed among the various detection methods (Supplementary Figure 5). The heterogeneity measures for both the combination of clinically and screen-detected cases (I^2 = 86.9%) and for clinically detected cases alone (I^2 = 85.9%) were lower than in the main analysis (Supplementary Table 3). This is likely due to the small number of studies included in these categories (n = 3 for both). In contrast, screen-detected showed even greater heterogeneity (I^2 = 98.8%).

Discussion

This systematic review and meta-analysis investigated the quantitative dose-response association between f-Hb concentrations in prior FIT and the detection of colorectal neoplasia. The findings showed a pooled effect size of 0.69 (95% CI, 0.58–0.79). Individuals with an f-Hb concentration of 10, 20, and 40 μg/g had a 5, 8, and 13 times higher risk of colorectal neoplasia at the subsequent screening round compared with individuals with 0 μg/g in their stool. Subgroup analyses suggested that the f-Hb concentration from a previous negative FIT was especially predictive for detecting AN in the following screening round.

Excluding the study by Senore et al[22] had the most significant impact on the pooled estimate. We hypothesize that this can be explained by the fact that Senore et al[22]

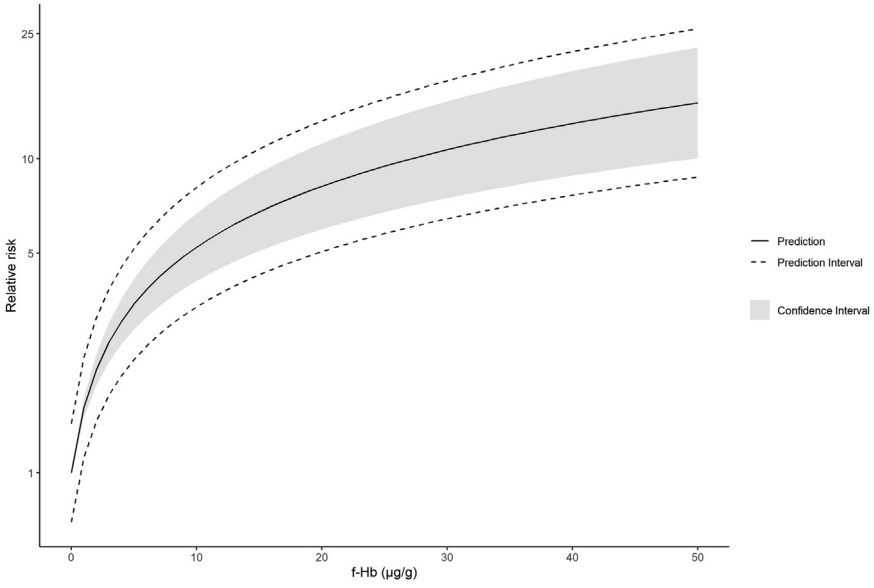


Figure 3. Predicted RR of detecting colorectal neoplasia based on prior f-Hb concentration.

considered the cumulative f-Hb from the 2 prior FITs instead of only the most recent FIT. If the f-Hb concentration increased between the first and the second round, averaging the f-Hb levels over both rounds will result in an average that is lower than the actual f-Hb concentration in the most recent round. Consequently, this underestimation of prior f-Hb in the most recent round might lead to incorrect RR assignments, leading to a steeper dose-response curve.

Despite the lack of statistical significance, studies that defined the outcome as any colorectal neoplasia generally reported smaller effect sizes than studies focusing exclusively on AN or CRC. This may be attributed to a lower sensitivity in detecting smaller, and nonadvanced adenomas. If small adenomas are present during previous FIT screenings but go undetected due to minimal or absence of bleeding, they may progress and become detectable in subsequent screenings. In addition, individuals are more likely to develop nonadvanced adenomas between 2 screening rounds compared with advanced adenomas or CRC. If more lesions are discovered in individuals with no or little f-Hb during previous screening rounds, the effect size will likely diminish.

The main strength of our study is its estimation of a dose-response association between f-Hb in prior FIT and the detection of colorectal neoplasia, with a large sample size of 4,493,672 participants. In contrast to previous research, which predominantly concentrated on specific geographic regions, this meta-analysis encompassed 13 studies conducted across multiple countries with a broad range of screening strategies, including variations in starting age, stopping age, and positivity cutoff. This broad scope enhances the generalizability of our findings, allowing us to estimate the increase in risk associated with f-Hb concentrations regardless of the differences between screening programs. Our results are relevant not only for countries where FIT is the primary CRC screening method but also for those like the United States, where colonoscopy is more commonly used. In these contexts, risk-tailored CRC screening intervals based on prior f-Hb levels could reduce the need for annual FIT testing, potentially making FIT a more appealing alternative to routine colonoscopy.

Nevertheless, several limitations should be acknowledged when interpreting our results, with some arising from limitations inherent in the existing literature rather than in our approach to its review. First, it is important to note that the broad inclusion of studies from various screening programs led to high heterogeneity in our meta-analyses, even after grouping similar studies in a subgroup analysis. A contributing factor to this heterogeneity is that, despite our efforts to standardize the data across different f-Hb categories and positivity cutoffs into a single pooled dose-response association, not all studies provided data for the first and second screening rounds. Variations in prior screening history, in which individuals were screened using different intervals and cutoffs, could influence the risk and therefore explain some of the observed heterogeneity. When heterogeneity is large, a random effects meta-analysis assigns relatively equal weights to studies compared with a fixed-effect analysis, potentially resulting in smaller studies exerting greater influence. If smaller studies report a stronger dose-response association, this could lead to an overestimation of the pooled effect size. However, we did not observe a consistent pattern between study size and effect size among the included studies.

Second, the lack of sufficient data posed challenges in exploring how other demographic variables might influence the risk estimate, or presenting results based on factors such as age and sex. Although all included studies that adjusted for age and sex found that both older age and male sex were associated with an increased risk for colorectal neoplasia in the next screening round, none of these studies examined the interaction between sex or age and f-Hb levels.

In addition, some variables extracted from the literature were too unbalanced to perform reliable subgroup analyses. For example, the OC-Sensor (Eiken Chemical Co) FIT brand was used in most of the studies, whereas the FOB-Gold (Sentinel) was used in only 3 Dutch studies. Given that both brands show equivalent accuracy in detecting AN,[28] we assume that the FIT brand does not affect the pooled RR significantly. However, differences in measurement sensitivity are present at the lower f-Hb concentrations, with the limits of blank, detection, and quantitation being 1 μg/g, 2 μg/g, and 3 μg/g, respectively, for the FOB-Gold, and 0, 1, and 6 μg/g for the OC-Sensor.[29] As a result, the variation in measurement sensitivity between the 2 brands may introduce inconsistencies in how reference categories are defined and could influence comparisons across studies.

Finally, the application of the Greenland and Longnecker[25] method to estimate correlations between the reference group and the risk at various exposure dose levels was hindered for certain studies in this meta-analysis due to the absence of available sample size and/or number of cases within each f-Hb category. This issue could potentially lead to an underestimation of the RR variance, which is a relevant concern as the pooled RR is determined by weighing each study's RR with the inverse of its variance. Nevertheless, a sensitivity analysis showed that our findings were robust even if we excluded studies with missing sample size and/or number of cases.

This research focused on asymptomatic patients. For symptomatic individuals, such as those experiencing changes in bowel habits, anemia, or other related symptoms, the distribution of f-Hb values may differ. Prior research has shown that symptomatic patients typically have higher f-Hb levels compared with asymptomatic individuals.[30] A different distribution of f-Hb concentrations suggests that the predicted RR based on f-Hb from prior screenings is not generalizable between symptomatic and asymptomatic individuals.

Our findings highlight the potential benefits of implementing risk-stratified screening based on prior f-Hb concentrations. Individuals with undetectable f-Hb levels have a significantly lower risk compared with those with detectable levels, suggesting that longer screening intervals may be appropriate for the former group and shorter intervals could be considered for those at higher risk. Such an approach could optimize resource allocation, reduce the

burden of unnecessary screenings, and maintain high detection rates for those at greater risk. This aligns with the framework proposed by O'Mahony et al,[31] who demonstrated the value of tailoring screening intervals to individual risk profiles using a mathematical model in breast cancer screening. Determining the optimal screening intervals within each national CRC screening program, however, requires further analysis. Successful implementation of risk-based screening requires consensus on the number of risk groups to define and whether additional risk factors, such as age and sex, should be incorporated. Striking a balance between the discriminatory power of a detailed risk classification and the practicality of implementation poses a significant challenge. Randomized controlled trials in the Netherlands and Italy are currently running, in which tailored invitation intervals are offered based on prior f-Hb concentrations. These studies hope to address some of these questions and provide practical insights into the real-world application of risk-stratified screening strategies.[32,33]

In addition, there are numerous possibilities to consider when tailoring screening strategies, including risk-group–specific invitation intervals, positivity cutoff levels, and different start and stop ages for screening. Future cost-effectiveness studies could help identify which risk-based approach offers the most favorable balance between benefits and potential harms. It is important to recognize that individuals with undetectable f-Hb levels make up most of those with a negative FIT result and ensure that reducing screening frequency for this low-risk group does not lead to unfavorable outcomes at the population level.

In conclusion, this systematic review and meta-analysis reveals that the risk of detecting colorectal neoplasia increases with f-Hb concentrations in prior FIT, but the rate of increase diminishes as f-Hb levels rise. These findings support the potential for risk-stratified CRC screening programs based on quantitative FIT values to improve the effectiveness of screening and optimize the harm-benefit ratio. However, we acknowledge the need for further investigation into the approach used to define risk groups, particularly in navigating the challenge of balancing comprehensive risk classification with practical implementation.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://doi.org/10.1053/j.gastro.2024.10.047.

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Conflicts of interest
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Data Availability
All meta-analyses reported in this study are available on request to the corresponding author, including data used for the analyses and analytic code.